Exp no. 1
Date: 20/7/26 Basic Networking commands

Aim:
Study of various network commands used in linux and windows.

1) arp - a
It shows the computer IP address and the IP and MAC address of devices on your local network.

Output:
Interface: 192.168.56.1 --- 0x7

Internet Address	Physical address	Type
192.168.56.255	ff-ff-ff-ff	static
224.0.0.2	01-00-5e-00-02	static

2) hostname:
It is the simplest of all TCP/IP commands

Output:
hp-hemalatha-ST

3) ip config /all:
It displays detailed configuration information about your TCP/IP connection including router, gateway, DNS, DHCP and type of ethernet adapter

Output: Windows IP configuration
Hostname ..... : hp-aldrine-11
Primary DNS Suffix
Node Type ..... : Mixed
IP Routing Enabled ..... : NO
WINS Proxy Enabled : NO

4. nbtstat - a
This command help solve problems with Net BIOS name resolution.

Output:-
Displays Protocol statistics and current TCP/IP connections using NBT
-a (adapter status) -c (cache)
-A (Adapter status) -n (names)

5) netstat:- Displays active network connections routing table and network statics

Proto	Local Address	Foreign Address	State
TCP	127.0.0.1.49678	hp_Aldrine_49679	Established

6) nslookup:- used to find DNS information that is IP address of a domain and DNS record

Output:-
Default server: Unknown
Address: IT2-16-8-1

7) Pathping:-
Unique to windows and combines the ping and Tracert commands
Output:-
Pathping [-g host_cost] [-h maximum-hops] [-i address] [-n] [-P period] [-q num]

8) Ping:- [Packet Internet groupes]
command is the host way to test connecting between two nodes.

LINUX Commands:

1) IP

This ip command is used to manage ip address network interfaces and routes.

output:

ip [OPTIONS] OBJECT { COMMAND | help }

ip [-force] -batch filename ||

2) ip config:

It was a state in many system admin tool for configuring and troubleshooting between network.

Output:

enp3s0: flags = 4163<OP, BROADCAST, RUNNING, MLTICAST > mtu 1500

inet 172.16.8.127 netmask 255.255.252.0

RX packets 335629 bytes

RX errors 0 dropped 17

3) mtr: MTR is a progress with command line interface that serves as a network diagnostic and troubleshooting tool.

output:

my traceroute [V0.87]

Host Packets

1.::1 Loss% snt last Avg Best Wost St.C

0.0% 145 0.2 0.1 0.1 0.6 0.C

4) tcpdump :-

The tcpdump command is designed for capturing and displaying packets.

Output :-

tcpdump: you don't have permission to capture on that device.

5) Ping :-

Ping is a tool that verifies IP-level connectivity to another TCP/IP computer by sending internet control Message.

Output :-

PING google.com (142.200.182.78) 56(84) bytes of data 64 bytes from lemma-aac-in
= 1 tt i = 112 time = 985 ms

Result :-

Thus the basic networking and Linux commands are learned.